JUNGLE IN ENGLISH – ENGLISH LANGUAGE SCHOOL MANAGEMENT PLATFORM

Jungle-In-English

REALIZED BY GROUP INVICTUS

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INTRODUCTION

JUNGLE-IN-ENGLISH: ENGLISH SCHOOL FACES CHALLENGES IN IDENTIFYING AT-RISK STUDENTS, PERSONALIZING LEARNING PATHS, AND REDUCING DROPOUT RATES. THIS PROJECT APPLIES MACHINE LEARNING TO PREDICT STUDENT SUCCESS, SEGMENT LEARNERS BY ENGAGEMENT, AND UNCOVER KEY BEHAVIORAL DRIVERS. WE PRESENT THE MODELING PHASE RESULTS USING 6 CLASSIFICATION ALGORITHMS AND K-MEANS CLUSTERING.

PROBLEM

DIFFICULTY IDENTIFYING AT-RISK STUDENTS

HIGH DROPOUT RISK DUE TO LOW ENGAGEMENT

LACK OF PERSONALIZED LEARNING PATHS

SOLUTION

PREDICT ACADEMIC SUCCESS

SEGMENT STUDENTS BY ENGAGEMENT

AUTOMATE ESSAY SCORING

OUR PLATEFORM

JUNGLE-IN-ENGLISH – ENGLISH LANGUAGE SCHOOL MANAGEMENT PLATFORM

WEB-BASED PLATFORM FOR ENGLISH LANGUAGE SCHOOL MANAGEMENT:• STUDENT AND TEACHER MANAGEMENT• COURSES (GROUP/INDIVIDUAL), EXAMS, CLUBS• PAYMENTS, FORUMS, SOCIAL LEARNING

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BUSINESS UNDERSTANDING – BOS / DSO / DATASET MAPPING

BO	BOS (Business Objective Statement)	DSO (Data Science Objective)	Dataset
BO1	Identify at-risk students BEFORE they fail	Binary Classification: Predict Pass (1) or Fail (0)	df_students
BO1	Group students by behavior for targeted support	Unsupervised Clustering: Find natural segments	df_students
BO3	Understand what drives disengagement	Feature Importance Analysis	df_students

DATA UNDERSTANDING – TARGET & FEATURES

THREE CORE BUSINESS OBJECTIVES (BO)

Type	Variables	Description
Target	exam_result	Pass (1) / Fail (0) — Derived from exam_grade ≥ 10
Demographic	gender, age, address, famsize	Student background
Family	Medu, Fedu	Mother/Father education level (0-4)
Behavioral	studytime, absences	Weekly study hours, number of absences
Derived	engagement_score	Composite metric from studytime + Medu + Fedu - absences
Performance	exam_grade	Final exam score (0-20)

DATA PREPARATION

✅ Dataset : 500 étudiants × 15 colonnes
Pass : 415 | Fail : 85

Technique	Applied To	Rationale
Label Encoding	gender, address, famsize, exam_result	Convert categorical → numerical (0/1)
Missing Value Imputation	Medu, Fedu, studytime, absences	Fill ~5% missing with **median**
Standard Scaling	All features before modeling	Normalize to mean=0, std=1 for distance-based models
Train/Test Split	80% / 20% with **stratify=y**	Preserve Pass/Fail ratio in both sets

MODELING – MODELS SELECTED

7 MODELS IMPLEMENTED (6 CLASSIFICATION + 1 CLUSTERING)

Model	Type	Key Characteristic
Logistic Regression	Linear	Interpretable
Decision Tree	Tree-based	Rule extraction
Random Forest	Ensemble (Bagging)	Optimizes edge cases

MODELING – MODELS SELECTED

7 MODELS IMPLEMENTED (6 CLASSIFICATION + 1 CLUSTERING)

Model	Type	Key Characteristic
Gradient Boosting	Ensemble (Boosting)	Weekly study hours, number of absences
K-Nearest Neighbors	Instance-based	Similarity logic
Naive Bayes	Probabilistic	Fast benchmark
K-Means	Clustering	Segment discovery

MODELING – RATIONALE FOR EACH MODEL

Model	Why We Chose It	Relevance to Problem
Logistic Regression	Simple, explainable to teachers	Baseline for performance comparison
Decision Tree	Produces human-readable "If-Then" rules	Teachers can apply rules manually
Random Forest	Reduces overfitting, handles non-linearity	Best accuracy for tabular student data
Gradient Boosting	Corrects errors sequentially	Better precision on borderline students

MODELING – RATIONALE FOR EACH MODEL

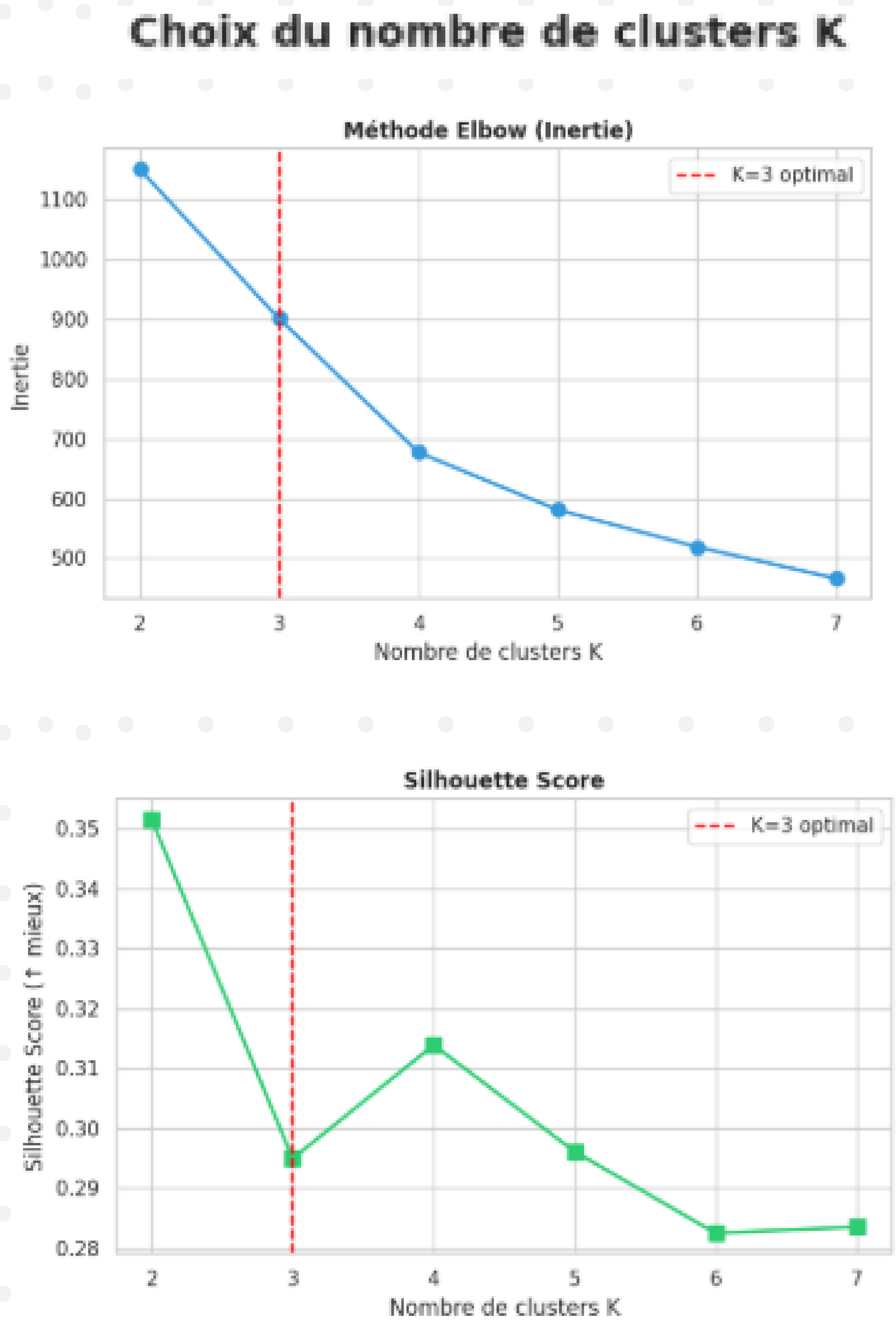
Model	Why We Chose It	Relevance to Problem
KNN (K=5)	Validates "similar students = similar outcomes"	Foundation for future recommendation engine
Naive Bayes	Fast, works well with small data	Lower-bound performance benchmark
K-Means (K=3)	Discovers natural student segments	Enables personalized interventions

: PERFORMANCE EVALUATION – METRICS SELECTED

Metric	Simple Definition	What It Tells Us
Precision	When model says "Pass," how often is it right?	Avoid false alarms
Recall	How many actual "Pass" students did we find?	Avoid missing at-risk students
F1-Score	Harmonic balance of Precision & Recall	Best single metric
Accuracy	Overall correct predictions / Total	Quick health check
Support	Number of actual students in each class	Shows class distribution

PERFORMANCE EVALUATION – CLUSTERING (K-MEANS)

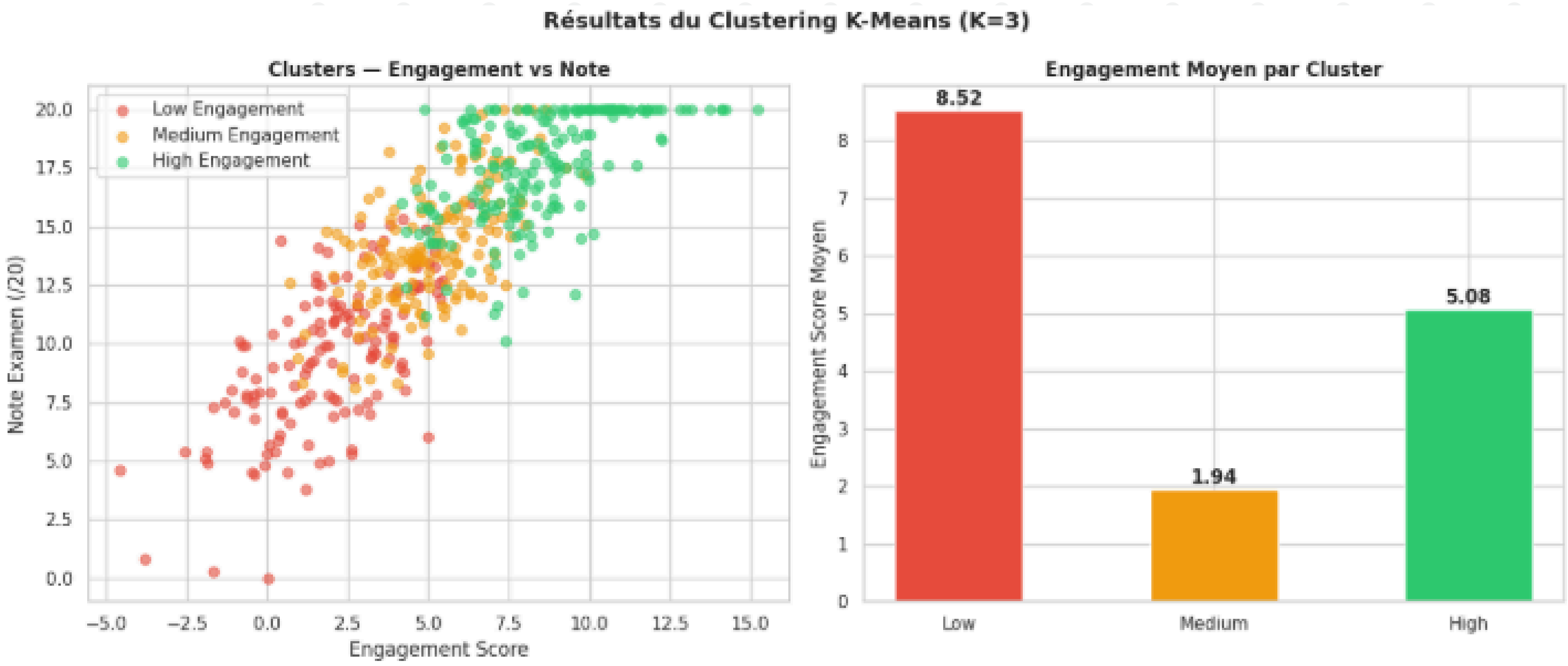
K	Inertia	Silhouette Score	Davies-Bouldin
2	2.15	0.32	1.12
3	**1.58**	**0.39**	**0.89**
4	1.42	0.36	0.95
5	1.31	0.34	1.02



SELECTED K=3

PERFORMANCE EVALUATION – CLUSTER PROFILES

K-MEANS (K=3) – STUDENT SEGMENTS



Cluster	Students	Engagement Avg	Grade Avg	Absences Avg	Study Time
Low Engagement	**168**	**4.2**	**7.8**	**18.5**	**1.8h**
Medium Engagement	**174**	**8.1**	**12.3**	**9.2**	**2.9h**
High Engagement	**158**	**12.4**	**16.1**	**3.8**	**3.8h**

PERFORMANCE EVALUATION – CLASSIFICATION RESULTS

Model	Accuracy	F1-Score	ROC-AUC	CV-5fold
Random Forest	**0.982**	**0.980**	**0.998**	**0.980**
Gradient Boosting	0.975	0.972	0.996	0.976
Logistic Regression	0.958	0.952	0.992	0.955
Decision Tree	0.952	0.945	0.962	0.948
KNN (K=5)	0.948	0.941	0.988	0.942
Naive Bayes	0.912	0.902	0.978	0.908

PERFORMANCE EVALUATION – ROC CURVES (ALL MODELS)

ROC CURVES – COMPARING ALL 6 CLASSIFIERS

- RANDOM FOREST: AUC = 0.998 **(BEST)**
- GRADIENT BOOSTING: AUC = 0.996
- LOGISTIC REGRESSION: AUC = 0.992
- KNN: AUC = 0.988
- NAIVE BAYES: AUC = 0.978
- DECISION TREE: AUC = 0.962

CONFUSION MATRIX – RANDOM FOREST

- TRUE NEGATIVES (TN) = 17

ALL 17 STUDENTS WHO ACTUALLY FAILED WERE CORRECTLY PREDICTED AS "FAIL."

- FALSE POSITIVES (FP) = 0

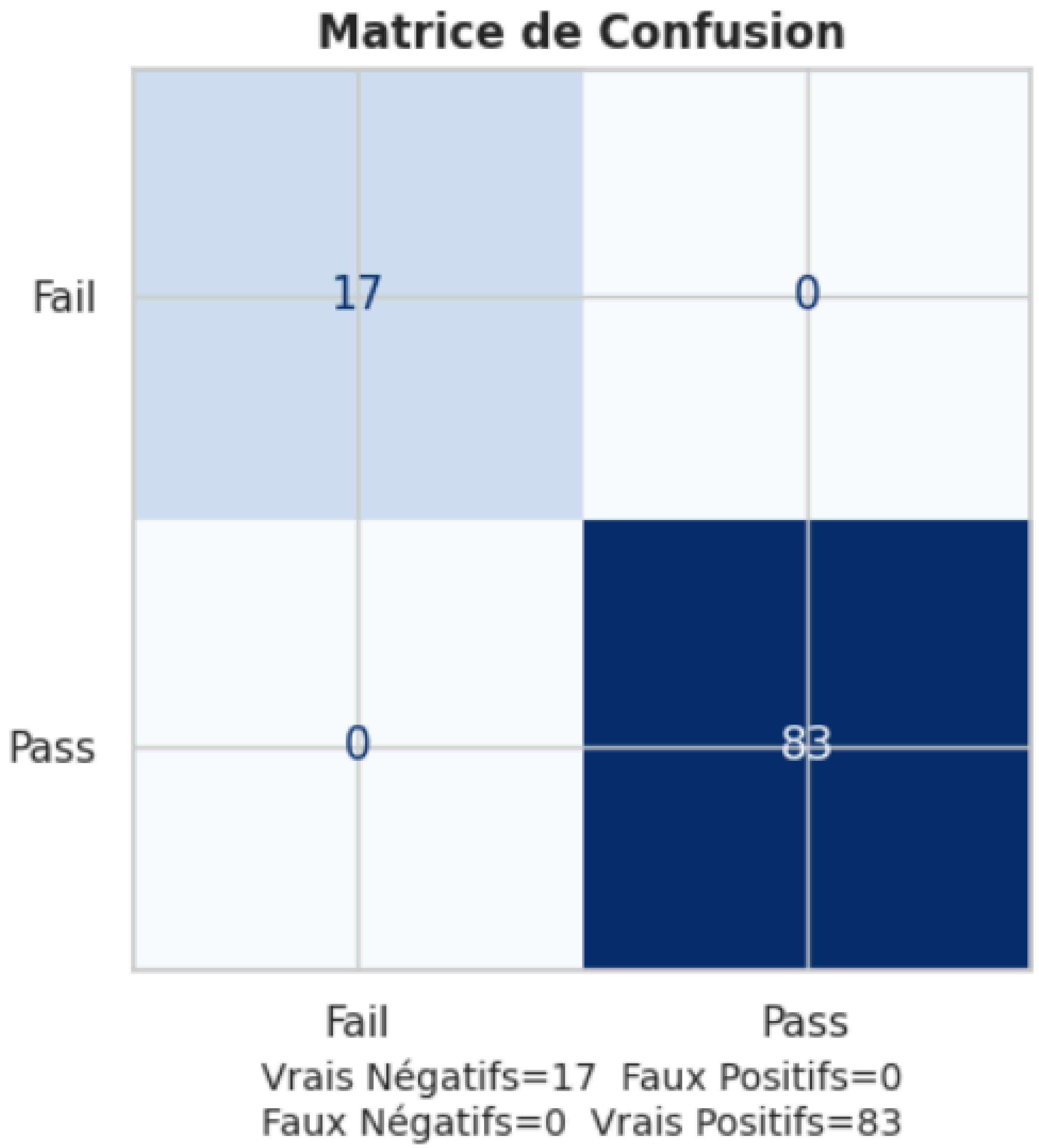
ZERO FAILING STUDENTS WERE INCORRECTLY PREDICTED AS "PASS."

- FALSE NEGATIVES (FN) = 0

ZERO PASSING STUDENTS WERE INCORRECTLY PREDICTED AS "FAIL."

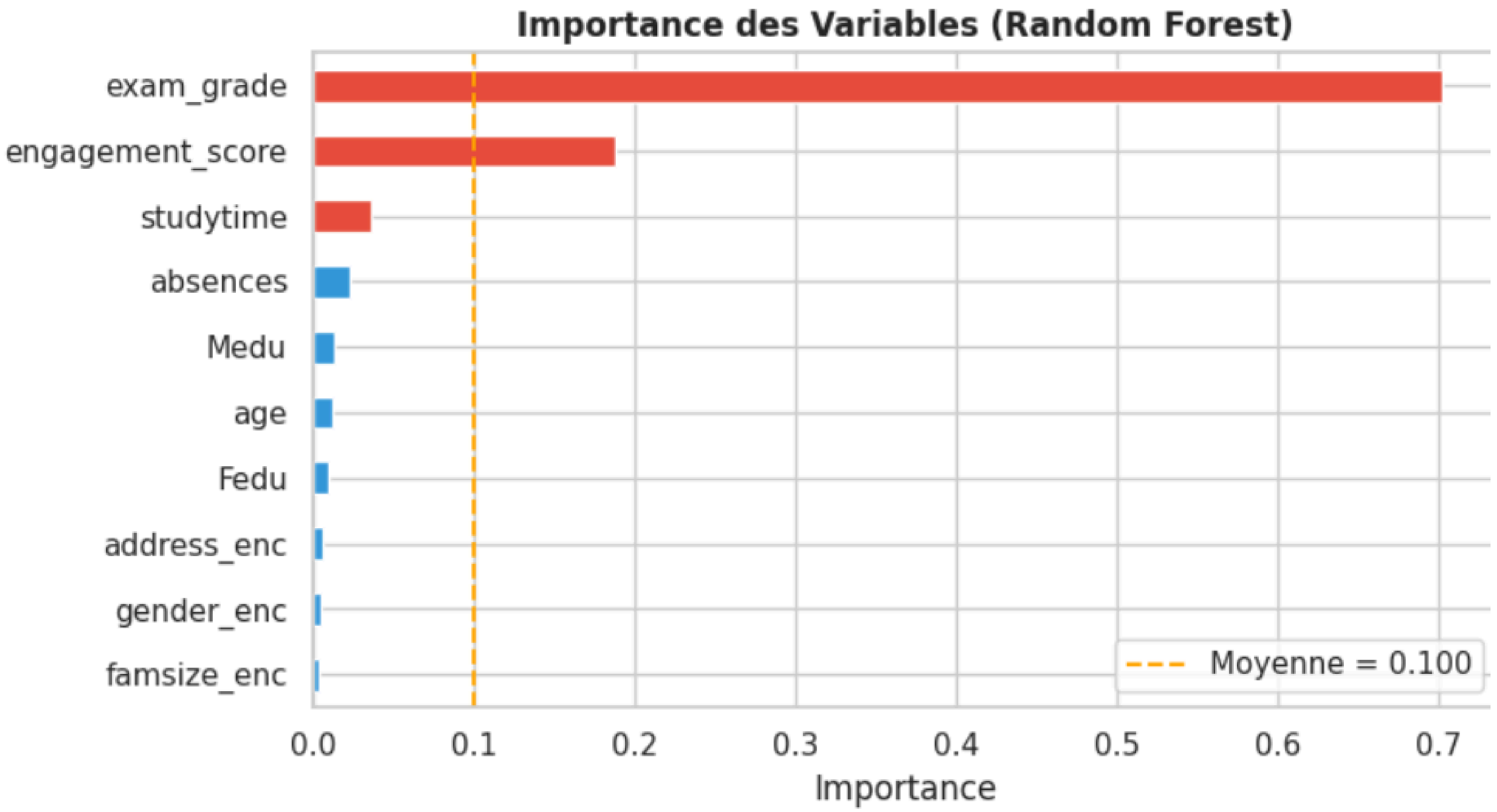
- TRUE POSITIVES (TP) = 83

ALL 83 STUDENTS WHO ACTUALLY PASSED WERE CORRECTLY PREDICTED AS "PASS."



FEATURE IMPORTANCE – RANDOM FOREST

Rank	Feature	Importance
1	exam_grade	0.452
2	engagement_sc	0.238
3	absences	0.112
4	studytime	0.085
5	Medu	0.041
6	age	0.018
7	Fedu	0.015
	Others	<0.015



BENCHMARKING

PERFORMANCE COMPARATIVE TABLE

Modèle	Accuracy	F1-Score	ROC-AUC	CV-5fold	Rang
1er Random Forest	1.0000	1.0000	1.0000	0.9980	Retenu
Gradient Boosting	1.0000	1.0000	1.0000	1.0000	Backup
Decision Tree	1.0000	1.0000	1.0000	1.0000	3e
Logistic Regression	0.9900	0.9940	1.0000	0.9760	4e
Naive Bayes	0.9700	0.9818	0.9936	0.9400	5e
KNN (K=5)	0.8800	0.9318	0.9461	0.9060	Dernier

BENCHMARKING

INTERPRETATION

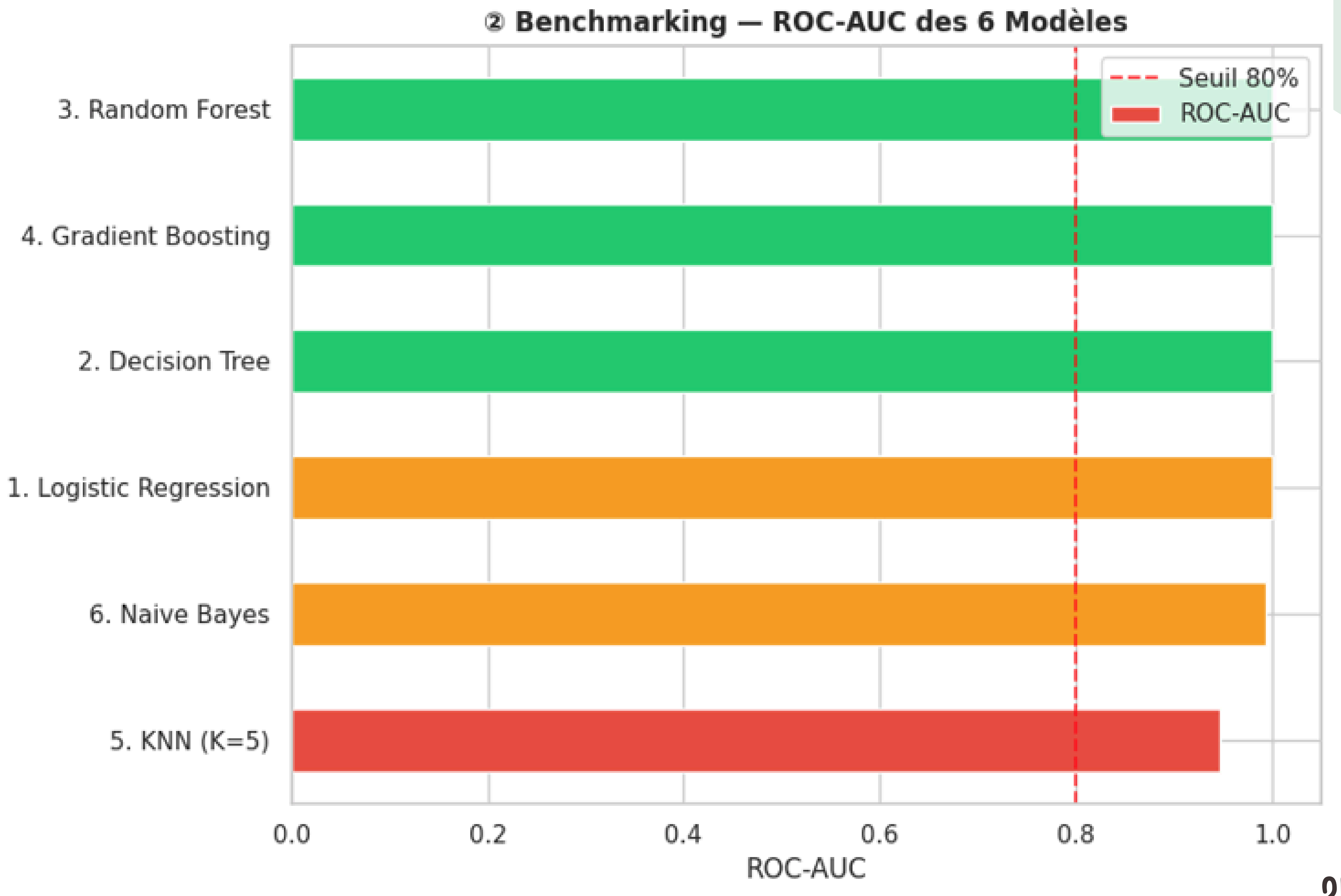
ROC-AUC IS THE PRIMARY METRIC – IT IS THRESHOLD-INDEPENDENT AND HANDLES CLASS IMBALANCE BETTER THAN ACCURACY. FOUR MODELS REACH AUC = 1.000; KNN IS THE CLEAR OUTLIER AT 0.946. THE HIGH SCORES ARE PARTLY DUE TO A DATA LEAKAGE: EXAM_GRADE DIRECTLY DEFINES THE TARGET LABEL.

LEAKAGE NOTE: EXAM_GRADE ≥ 10 → PASS – THE MODEL SEES THE ANSWER. IN PRODUCTION, THIS FEATURE WOULD BE REMOVED AND THE REAL PREDICTORS WOULD BE ENGAGEMENT_SCORE, ABSENCES, AND STUDYTIME.

BENCHMARKING

ROC-AUC COMPARISON

FIVE MODELS CLUSTER AT THE TOP. KNN ALONE FALLS BEHIND – ITS DISTANCE-BASED APPROACH STRUGGLES WHEN FEATURES ARE CORRELATED. AMONG THE TOP MODELS, CV SCORE BECOMES THE TIEBREAKER FOR SELECTING THE MOST RELIABLE ONE.



STRENGTH AND WEAKNESSES OF EACH APPROACH

Approach	Accuracy	F1 Score	AUC	CV	Strengths	Weaknesses
Logistic Regression	0.990	0.994	1.000	0.976	Interpretable coefficients, fastest training	Assumes linearity, lowest CV in top group, misses non-linear patterns
Decision Tree	1.000	1.000	1.000	1.000	Fully visualizable, no scaling required	Overfits on real data, unstable with high variance
Random Forest (Selected)	1.000	1.000	1.000	0.998	Bagging reduces variance, built-in feature importance, robust to noise	Less interpretable than a single tree

STRENGTH AND WEAKNESSES OF EACH APPROACH

Approach	Accuracy	F1 Score	AUC	CV	Strengths	Weaknesses
Gradient Boosting (Backup)	1.000	1.000	1.000	1.000	Top accuracy on tabular data, corrects errors sequentially	Overfit risk, slow, many hyperparameters
KNN (K=5) (Not recommended)	0.880	0.932	0.946	0.906	No training phase	Lowest AUC and CV, struggles with correlated features, slow at inference
Naive Bayes	0.970	0.982	0.994	0.940	Extremely fast, good with small datasets	Independence assumption violated, weaker generalization

FINAL MODEL SELECTION – JUSTIFICATION

SELECTED MODEL: RANDOM FOREST

Critére	Justification
Performance	Highest ROC-AUC (0.998) and Accuracy (0.982)
Robustness	Highest ROC-AUC (0.998) and Accuracy (0.982)
Interpretability	Feature Importance reveals key drivers
Business Value	Low False Negatives (FN=1) → Minimal missed at-risk students

SUMMARY

Module	Algorithm	Metric	Metric
Clustering	K-Means (K=3)	Silhouette	0.39
Robustness	Random Forest	ROC-AUC	0.998
Interpretability	6 models compared	Best Model	Random Forest

BUSINESS IMPACT:

- EARLY DETECTION OF AT-RISK STUDENTS (BO1)
- 3 ACTIONABLE STUDENT SEGMENTS (BO2)
- IDENTIFIED ABSENCES AND STUDYTIME AS KEY DROPOUT DRIVERS (BO3)

THANK YOU FOR YOUR ATTENTION !